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39. Given $2Al + Fe_2O_3 \rightarrow Al_2O_3 + 2Fe$

	Reactants		Products	
	Al	**Fe_2O_3**	**Al_2O_3**	**Fe**
Amount of reactant allowed to react	324 g	1.12 kg	-	-
Number of moles allowed to react	$\frac{324}{27} = 12$ mol	$\frac{1.12 \times 10^3}{160} = 7$ mol	-	-
Stoichiometric Co-efficient	2	1	1	2
Number of moles consumed during reaction	12 mol	6 mol	-	-
Number of moles of reactant unreacted and number of moles of product formed	-	1 mol	6 mol	12 mol

Molar mass of Al_2O_3 formed

$= 6 \text{ mol} \times 102 \text{ g mol}^{-1}$ $\left[\begin{array}{l} Al_2O_3 \\ (2 \times 27) + 3(\times 16) \\ 54 + 48 = 102 \end{array}\right]$ $= 612$ g

Excess reagent = Fe_2O_3

Amount of excess reagent left at the end of the reaction = 1 mol × 160 g mol^{-1}

$= 160$ g $\left[\begin{array}{l} Fe_2O_3 \\ (2 \times 56) + (3 \times 16) \\ 112 + 48 = 160 \end{array}\right]$ $= 160$ g

40. Balanced equation for the combustion of ethane

$$C_2H_6 + \tfrac{7}{2} O_2 \rightarrow 2CO_2 + 3H_2O$$

$$\Rightarrow \quad 2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6\ H_2O$$

To produce 4 moles of CO_2, 2 moles of ethane is required

$\therefore$ To produce 1 mole (44 g) of CO_2 required